

计算物理 B 公式汇总

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1. 蒙特卡洛方法

1.1 随机数与随机变量

蒙特卡洛方法：

$$|I_n - I| < \lambda \frac{\sigma}{\sqrt{n}} \quad (1)$$

生成随机数：

$$x_{n+1} = [10^{-r} x_n^2] \bmod (10^{2r}), \quad \xi_n = \frac{x_n}{10^{2r}} \in [0, 1]. \quad (2)$$

或者

$$x_{n+1} = (ax_n + c) \bmod m, \quad \xi_n = \frac{x_n}{m}. \quad (3)$$

随机数的检验：

$$\text{均匀性: } \chi^2 = \sum_{i=1}^k \frac{(N_i - N/k)^2}{N/k} \sim \chi^2(k-1). \quad (4)$$

$$\text{独立性: } \chi^2 = \sum_{i,j=1}^k \frac{(N_{i,j} - N/k^2)^2}{N/k^2} \sim \chi^2((k-1)^2). \quad (5)$$

$\chi^2 < t_a$ 则通过检验.

1.2 初等抽样

离散变量：

$$x = x_j, \quad \text{if } F(x_{j-1}) \leq \xi < F(x_j). \quad (6)$$

反函数法：

$$\xi \sim U[0, 1], \quad x = F^{-1}(\xi). \quad (7)$$

变换抽样（例子）：

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2} \quad (8)$$

从而

$$\begin{cases} x = \sqrt{-2 \ln u} \cos(2\pi v), \\ y = \sqrt{-2 \ln u} \sin(2\pi v), \end{cases} \quad u, v \sim U[0, 1]. \quad (9)$$

1.3 舍选法

1.3.1 第 I 类舍选法

$$x \in [a, b] \times [0, L], \quad L = \max f(x). \quad (10)$$

抽 $\xi_1, \xi_2 \sim U[0, 1]$, 若

$$\xi_2 \leq \frac{f(\xi_1)}{L}, \quad (11)$$

则保留 ξ_1 , 效率为

$$\eta = \frac{1}{L(b-a)}. \quad (12)$$

1.3.2 第 II 类舍选法

$$f(x) = L \frac{f(x)}{Lg(x)} g(x), \quad L = \max \left[\frac{f(x)}{g(x)} \right]. \quad (13)$$

要求 $f(x), g(x)$ 为概率分布,

$$h = \frac{f(x)}{Lg(x)} \in [0, 1]. \quad (14)$$

抽 $\xi_1 \sim g(x)$, $\xi_2 \sim U[0, 1]$, 若

$$\xi_2 \leq \frac{1}{L} \frac{f(\xi_1)}{g(\xi_1)}, \quad (15)$$

则保留 ξ_1 , 效率为

$$\eta = \frac{1}{L}. \quad (16)$$

1.3.3 第 III 类舍选法

$$f(x) = L \int_{-\infty}^{h(x)} g(x, y) dy, \quad g(x, y) \text{ 为分布}. \quad (17)$$

抽 $\xi_1, \xi_2 \sim g(x, y)$, 若

$$\xi_2 \leq h(\xi_1), \quad (18)$$

则保留 ξ_1 , 效率为

$$\eta = \frac{1}{L} = \int_{-\infty}^{+\infty} \int_{-\infty}^{h(x)} g(x, y) dx dy. \quad (19)$$

以 $h(x)$ 包住的点多为优.

如 $\theta \in [0, \pi]$ 的余弦分布:

$$f(x) = \frac{1}{\pi\sqrt{1-x^2}}, \quad |x| < 1. \quad (20)$$

抽 $\xi_1, \xi_2 \sim U[0, 1]$, 若

$$\xi_1^2 + \xi_2^2 \leq 1, \quad (21)$$

则保留

$$\frac{\xi_1^2 - \xi_2^2}{\xi_1^2 + \xi_2^2}. \quad (22)$$

1.4 复合抽样

连续:

$$f(x) = \int_{-\infty}^{+\infty} g(x|y)h(y) dy. \quad (23)$$

按照 $h(y)$ 抽一个 y_0 , 再根据 $g(x|y_0)$ 抽 x .

离散 (加分布抽样):

$$f(x) = \sum_i p_i g_i(x), \quad p_i \text{ 和为 } 1, \quad g_i(x) \text{ 为密度}. \quad (24)$$

抽 $\xi \sim U[0, 1]$, 若

$$\sum_{i=1}^{n-1} p_i < \xi \leq \sum_{i=1}^n p_i, \quad (25)$$

则抽一个 $g_n(x)$ 保留。

1.5 幂次抽样

$$f(x) = nx^{n-1}, \quad x \in [0, 1]. \quad (26)$$

抽 $\xi_1, \xi_2, \dots, \xi_n \sim U[0, 1]$, 保留 $\max(\xi_1, \xi_2, \dots, \xi_n)$.

1.6 其他抽样方法与技巧

平移法: 自动调整归一化系数

例: $r \in [R_0, R_1]$, 则根据 $r = (R_1 - R_0)x + R_0, x \in [0, 1]$, 先在 $[0, 1]$ 上抽 x 再代入. 实际上是某种一维变换.

乘加分布抽样: 化为加分布抽样

$$f(x) = H_1(x)g_1(x) + H_2(x)g_2(x) = p_1 g_1^{\text{eff}}(x) + p_2 g_2^{\text{eff}}(x). \quad (27)$$

其中:

$$p_i = \int_a^b H_i(x)g_i(x) dx \quad (28)$$

减分布抽样:

$$f(x) = a_1 g_1(x) - a_2 g_2(x) \quad (29)$$

$$\Rightarrow f(x) = (a_1 - a_2 m) \frac{f(x)}{g_1(x)(a_1 - a_2 m)} g_1(x), \quad m = \left[\frac{g_2(x)}{g_1(x)} \right]_{\min}. \quad (30)$$

因此：（第二类舍选法）

$$0 < f(x) = g_1(x) \left(a_1 - a_2 \frac{g_2(x)}{g_1(x)} \right) \leq (a_1 - a_2 m) g_1(x). \quad (31)$$

$$\eta = \frac{1}{a_1 - a_2 m}. \quad (32)$$

乘减分布抽样：

$$f(x) = H_1(x)g_1(x) - H_2(x)g_2(x) = H_1(x)g_1(x) \left(1 - \frac{H_2 g_2}{H_1 g_1} \right) \leq M g_1(x)(1 - m). \quad (33)$$

因此：（第二类舍选法）

$$f(x) = M(1 - m) \frac{f(x)}{M g_1(x)(1 - m)} g_1(x), \quad \eta = \frac{1}{M(1 - m)}. \quad (34)$$

乘加（另一种看法）：

$$f(x) \leq g_1(x) \left(H_1(x) + \frac{H_2(x)g_2(x)}{g_1(x)} \right) \leq g_1(x) M_{\text{eff}}. \quad (35)$$

$$f(x) = M_{\text{eff}} \frac{f(x)}{M_{\text{eff}} g_1(x)} g_1(x), \quad \eta = \frac{1}{M_{\text{eff}}}. \quad (36)$$

（第二类舍选法）

1.7 高维舍选法

$$\vec{\eta} = (\eta_1, \dots, \eta_n)^T \sim f(x_1, x_2, \dots, x_n), \quad (37)$$

$$x_i \in [a_i, b_i], \quad L = \max f(x_1, x_2, \dots, x_n). \quad (38)$$

抽 $\xi_1, \xi_2, \dots, \xi_{n+1} \sim U[0, 1]$. 若

$$\xi_{n+1} \leq \frac{1}{L} f((b_1 - a_1)\xi_1 + a_1, \dots, (b_n - a_n)\xi_n + a_n), \quad (39)$$

则保留 $\eta = (\xi_1, \xi_2, \dots, \xi_n)$, 效率为

$$\eta_{\text{eff}} = \frac{1}{L \prod_i (b_i - a_i)}. \quad (40)$$

1.8 条件抽样法

$$f(x_1, \dots, x_n) = f_n(x_n | x_1, \dots, x_{n-1}) f_{n-1}(x_{n-1} | x_1, \dots, x_{n-2}) \cdots f_2(x_2 | x_1) f_1(x_1). \quad (41)$$

由 $f_1(x_1)$ 抽 x_1 , 再由 $f_2(x_2 | x_1)$ 抽 x_2 , 一直到 x_n , 完成一组抽样。

1.9 蒙特卡洛计算积分的方法

- **分层抽样：** 极小值少抽样、极大值多抽样。
- **重要抽样：** 找 $[a, b]$ 上归一分布 $g(x)$, $[a, b]$ 上抽满足 $g(x)$ 的 ξ 得

$$\bar{I} = \frac{1}{N} \sum_{i=1}^N \frac{f(\xi_i)}{g(\xi_i)}. \quad (42)$$

- **简单均匀抽样：**

$$\xi \sim U[a, b], \quad I = (a - b) \frac{1}{N} \sum_{i=1}^N f(\xi_i). \quad (43)$$

1.10 重要抽样常见错误分析 * (例)

$$\int_7^{12} f(x) e^{-x} dx. \quad (44)$$

(用 Metropolis 方法) 生成了 $[7, 12]$ 的 e^{-x} 抽样结果, 问

$$\frac{\sum f(\xi_i)}{N} \quad (45)$$

与 I 什么关系. 分析如下:

$$I = \int_7^{12} \frac{f(x) e^{-x}}{C e^{-x}} C e^{-x} dx. \quad (46)$$

则积分值的正确估计结果为

$$\frac{1}{N} \sum_{i=1}^N \frac{f(\xi_i)}{C}. \quad (47)$$

又

$$\frac{1}{C} = \int_7^{12} e^{-x} dx, \quad (48)$$

从而

$$I = \int_7^{12} e^{-x} dx \left(\frac{1}{N} \sum_{i=1}^N f(\xi_i) \right). \quad (49)$$

二者并不相等!

1.11 控制变量与对偶变量法

控制变量法：

$$I = \int_0^1 f(x) dx = \int_0^1 [f(x) - g(x)] dx + \int_0^1 g(x) dx. \quad (50)$$

对偶变量法：

$$I = (a - b) \frac{1}{N} \sum_{i=1}^N f(\xi_i) = (a - b) \frac{1}{N} \sum_{i=1}^N \frac{f(\xi_i) + f(a + b - \xi_i)}{2}. \quad (51)$$

1.12 随机游走

抽 $\xi_1, \dots, \xi_N \sim U[0, 1]$. 若 $\xi_i < p$ 则向左, 否则向右。

$$x = \sum_{i=1}^N x_i. \quad (52)$$

一、二维回原点概率 $P = 1$, 三维 $P = 34\%$ 。

1.13 解 Poisson 方程

$$\phi_0 = \frac{1}{4}(\phi_1 + \phi_2 + \phi_3 + \phi_4 - h^2 q_0). \quad (53)$$

$$\bar{\phi}_0 = \frac{1}{N} \sum_{i=1}^N \eta_0^{(i)} = \frac{1}{N} \sum_{i=1}^N \left(F(s^{(i)}) - \frac{h^2}{4} \sum_j q_j^{(i)} \right). \quad (54)$$

问 4 个方向的均等概率遵循随机游走。

1.14 Metropolis 方法抽 $f(x)$

- 选取初始点 x_0 , n 到 $n+1$ 步如下操作:

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$$x_{\text{try}} = x_n + \eta_n, \quad \eta_n \in [-\delta, +\delta]. \quad (55)$$

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$$r = \frac{f(x_{\text{try}})}{f(x_n)}. \quad (56)$$

- 若 $r \geq 1$, 则

$$x_{n+1} = x_{\text{try}}. \quad (57)$$

否则生成 $\xi \sim U[0, 1]$, 若 $\xi < r$, 则 $x_{n+1} = x_{\text{try}}$, 否则 $x_{n+1} = x_n$ 。

- 继续试探 x_{n+2} .

1.15 粒子物理模拟

透射率:

$$\eta_i = \begin{cases} 1, & \text{if } x_i > a, \\ 0, & \text{else.} \end{cases} \quad (58)$$

$$\frac{\eta}{N} = \frac{1}{N} \sum_{i=1}^N \eta_i. \quad (59)$$

方差:

$$\frac{1}{N} \text{Var}(i) = \frac{1}{N} \left(1 - \frac{N_{\text{acc}}}{N} \right) \frac{N_{\text{acc}}}{N}. \quad (60)$$

1.16 统计物理模拟

$$\langle A \rangle = \int A(\vec{q}_i, \vec{p}_i) \rho(\vec{q}_i, \vec{p}_i) d\Omega. \quad (61)$$

使用态平均求宏观量.

求均值:

$$\langle A \rangle = \frac{1}{Z} \int A(\vec{x}) e^{-E(\vec{x})/k_B T} d\vec{x}. \quad (62)$$

利用 $\eta \sim f/Z$ 抽 m 个样本:

$$\langle A \rangle = \frac{1}{m} \sum_i A(\eta_i). \quad (63)$$

Metropolis 方法:

$$w = \min \left[1, \exp \left(\frac{V(x_n)}{k_B T} - \frac{V(x_{\text{try}})}{k_B T} \right) \right]. \quad (64)$$

e.g. 伊辛模型: 相当于先形成一组 Boltzmann 分布的态, 每次随机游走以 $1/N$ 的概率翻转.

$$w = \min \left[1, \exp \left(\frac{H(s_n)}{k_B T} - \frac{H(s_{\text{try}})}{k_B T} \right) \right]. \quad (65)$$

$$\langle M \rangle = \frac{1}{n} \sum_{i=m+1}^{m+n} M(s_i), \quad M = \sum_{i=1}^N s_i. \quad (66)$$

2. 机器学习

2.1 损失函数与线性回归

$$J(y, f) = \sum_i |y - f_i|^2. \quad (67)$$

线性回归:

$$J(\vec{w}, b) = \frac{1}{2m} \sum_i \left(\vec{w} \cdot \vec{x}^{(i)} + b - y^{(i)} \right)^2. \quad (68)$$

$$f_{\vec{w}, b}(\vec{x}^{(i)}) = \vec{w} \cdot \vec{x}^{(i)} + b. \quad (69)$$

2.2 梯度下降

$$w_{\text{next}} = w - \alpha \frac{\partial J(w, b)}{\partial w}, \quad b_{\text{next}} = b - \alpha \frac{\partial J(w, b)}{\partial b}. \quad (70)$$

α 称为学习率.

$$w'_j = w_j - \alpha \frac{\partial J(w, b)}{\partial w_j} = w_j - \frac{\alpha}{m} \sum_{i=1}^m \left(\vec{w} \cdot \vec{x}^{(i)} + b - y^{(i)} \right) x_j^{(i)}. \quad (71)$$

$$b' = b - \alpha \frac{\partial J(w, b)}{\partial b} = b - \frac{\alpha}{m} \sum_{i=1}^m \left(\vec{w} \cdot \vec{x}^{(i)} + b - y^{(i)} \right). \quad (72)$$

2.3 逻辑回归

$$g(z) = \frac{1}{1 + e^{-z}}, \quad f_{\vec{w}, b}(\vec{x}) = g(\vec{w} \cdot \vec{x} + b). \quad (73)$$

损失: Binary cross entropy:

$$L = -y^{(i)} \log f_{\vec{w}, b}(x^{(i)}) - (1 - y^{(i)}) \log (1 - f_{\vec{w}, b}(x^{(i)})). \quad (74)$$

代价函数:

$$J = \frac{1}{m} \sum_{i=1}^m \left[-y^{(i)} \log f_{\vec{w}, b}(x^{(i)}) - (1 - y^{(i)}) \log (1 - f_{\vec{w}, b}(x^{(i)})) \right]. \quad (75)$$

2.4 正则化

正则化代价函数:

$$\tilde{J} = \frac{1}{2m} \sum_{i=1}^m \left(\vec{w} \cdot \vec{x}^{(i)} + b - y^{(i)} \right)^2 + \frac{\lambda}{2m} \sum_{j=1}^n w_j^2. \quad (76)$$

对所有 w_j .

梯度下降:

$$w'_j = w_j - \frac{\alpha \lambda}{m} w_j + \dots = \left(1 - \frac{\alpha \lambda}{m} \right) w_j + \dots. \quad (77)$$

$$\text{Lasso:} \quad + \lambda \sum_{j=1}^n |w_j|, \quad \text{Ridge:} \quad + \lambda \sum_{j=1}^n w_j^2. \quad (78)$$

2.5 评价指标

决定系数:

$$R^2 = 1 - \frac{\sum_{i=1}^m (y_i - \hat{y}_i)^2}{\sum_{i=1}^m (y_i - \bar{y})^2}. \quad (79)$$

$R^2 = 1$ 则优, $R^2 = 0$ 则与均值差不多。

精确率:

$$\text{precision} = \frac{\text{True pos}}{\text{True pos} + \text{False pos}}. \quad (80)$$

召回率:

$$\text{Recall} = \frac{\text{True pos}}{\text{True pos} + \text{False neg}}. \quad (81)$$

$$F1 \text{ score} = 2 \frac{PR}{P + R}. \quad (82)$$

足够置信度才能判断的：精确率高（阈值高）；不能漏过一个的检查：召回率高（阈值低）。

2.6 神经网络前向传播

$$\mathbf{a}^{(1)} = \begin{pmatrix} g(\mathbf{w}_1^{(1)} \cdot \mathbf{x} + b_1^{(1)}) \\ \vdots \\ g(\mathbf{w}_f^{(1)} \cdot \mathbf{x} + b_f^{(1)}) \end{pmatrix}, \quad \mathbf{a}^{(2)} = \begin{pmatrix} g(\mathbf{w}_1^{(2)} \cdot \mathbf{a}^{(1)} + b_1^{(2)}) \\ \vdots \\ g(\mathbf{w}_f^{(2)} \cdot \mathbf{a}^{(1)} + b_f^{(2)}) \end{pmatrix}, \quad \mathbf{a}^{(3)} = g(\mathbf{w}_1^{(3)} \cdot \mathbf{a}^{(2)} + b_1^{(3)}). \quad (83)$$

2.7 反向传播

$$\frac{\partial J}{\partial w^{(l)}} = \frac{\partial J}{\partial a^{(l)}} \frac{\partial a^{(l)}}{\partial z^{(l)}} \frac{\partial z^{(l)}}{\partial w^{(l)}} = (a^{(l)} - y)g'(z^{(l)})a^{(l-1)}. \quad (84)$$

$$\frac{\partial J}{\partial b^{(l)}} = \frac{\partial J}{\partial a^{(l)}} \frac{\partial a^{(l)}}{\partial z^{(l)}} \cdot 1. \quad (85)$$

$$J = \frac{1}{2}(a^{(l)} - y)^2, \quad a^{(l)} = g(z^{(l)}), \quad z^{(l)} = w^{(l)}a^{(l-1)} + b^{(l)}. \quad (86)$$

$$\frac{\partial J}{\partial w^{(l-1)}} = \frac{\partial J}{\partial a^{(l-1)}} \frac{\partial a^{(l-1)}}{\partial z^{(l-1)}} \frac{\partial z^{(l-1)}}{\partial w^{(l-1)}}. \quad (87)$$

$$\frac{\partial J}{\partial a^{(l-1)}} = \frac{\partial J}{\partial a^{(l)}} \frac{\partial a^{(l)}}{\partial z^{(l)}} \frac{\partial z^{(l)}}{\partial a^{(l-1)}}. \quad (88)$$

2.8 Softmax 多分类问题

$$a_i = \frac{e^{z_i}}{e^{z_1} + e^{z_2} + \dots + e^{z_{10}}} = P(y = i | \vec{x}). \quad (89)$$

$$z_i^{(3)} = \mathbf{w}_i^{(3)} \cdot \mathbf{a}^{(2)} + b_i^{(3)}. \quad (90)$$

$$\text{loss}(a_1, a_2, \dots, a_n; y) = - \sum_{j=1}^n p_j \log a_j = - \log a_i \text{ (if } y_i = i). \quad (91)$$

2.9 决策树

信息熵:

$$H(p_1) = -p_1 \log_2 p_1 - p_0 \log_2 p_0, \quad p_0 = 1 - p_1. \quad (92)$$

$H(p_1)$ 小说明纯度高.

信息增益:

$$\text{information gain} = H(p_1^{\text{root}}) - (w_L H_L(p_1) + w_R H_R(p_1)). \quad (93)$$

目标是最大化样本纯度.

方差减少:

$$\text{Var}(\text{root}) - w_L \text{Var}(\text{left}) - w_R \text{Var}(\text{right}). \quad (94)$$

3. 分子动力学

3.1 Verlet 方法

二步 Verlet 方法: .

$$x(t+h) = 2x(t) - x(t-h) + h^2 \frac{F(x)}{m}. \quad O(h^4) \quad (95)$$

$$p(t) = m \frac{x(t+h) - x(t-h)}{2h}. \quad O(h^2) \quad \text{中心差分} \quad (96)$$

速度落后一步, 无法自启动 (给出前两步位置) .

Leap-frog:

$$x(t+h) = x(t) + v\left(t + \frac{h}{2}\right)h, \quad (97)$$

$$v\left(t + \frac{h}{2}\right) = v\left(t - \frac{h}{2}\right) + a(t)h. \quad (98)$$

速度落后半步.

Verlet 速度形式:

$$x(t+h) = x(t) + v(t)h + \frac{h^2}{2}a(t), \quad (99)$$

$$v(t+h) = v(t) + \frac{a(t) + a(t+h)}{2}h. \quad (100)$$

同步更新.

3.2 L-J 势能

$$V_{ij}(r_{ij}) = 4\varepsilon_{ij} \left[\left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}} \right)^6 \right]. \quad (101)$$

$$F_{ix} = 48 \frac{\varepsilon}{\sigma^2} \sum_{j=1(\neq i)}^N (x_i - x_j) \left[\left(\frac{\sigma}{r} \right)^{14} - \frac{1}{2} \left(\frac{\sigma}{r} \right)^8 \right]. \quad (102)$$

周期性边界条件:

$$r(t+h) \rightarrow r(t+h) \pm L. \quad (103)$$

最小像力约定:

$$r_{ij} = \min |\vec{r}_i - \vec{r}_j + nL| < r_c. \quad (104)$$

3.3 宏观物理量的读取

动能 (从平衡态的 n_0 步计算到第 n 步.)

$$\bar{E}_k = \frac{1}{n - n_0} \sum_{\mu > n_0}^n \sum_{i=1}^N \frac{(p_i^2)^{(\mu)}}{2m}. \quad (105)$$

温度、势能

$$\bar{E}_k = \frac{d}{2} N k_B T, \quad d \text{ 为单粒子自由度, 无约束系统,} \quad (106)$$

$$\bar{U} = \frac{1}{n - n_0} \sum_{\mu > n_0}^n \sum_{i < j} (u(r_{ij}))^{(\mu)} + U_c. \quad (107)$$

长程修正项:

$$U_c = N \rho 2\pi \int_{r_c}^{\infty} u(r) g(r) r^2 dr. \quad (108)$$

$$U_{\text{total}} = \frac{N}{2} \int_0^{\infty} u(r) \rho g(r) 4\pi r^2 dr. \quad (109)$$

L-J 势能长程修正 (取 $g(r) = 1$):

$$U_c = \frac{8\pi N \rho \varepsilon \sigma^3}{3} \left[\frac{1}{3} \left(\frac{\sigma}{r_c} \right)^9 - \left(\frac{\sigma}{r_c} \right)^3 \right]. \quad (110)$$

3.4 对关联函数

$$\rho(\vec{r}) \simeq \frac{\langle N(\vec{r}, \Delta\vec{r}) \rangle}{\Omega(\vec{r}, \Delta\vec{r})}, \quad \text{时间平均的粒子数密度,} \quad \Omega = 4\pi r^2 \Delta r. \quad (111)$$

$$g(\vec{r}) \simeq \frac{\Delta N(\vec{r}, \Delta\vec{r})}{\rho 4\pi r^2 \Delta r}. \quad (112)$$

粒子数密度随粒子对相对位矢的变化.

3.5 速度修正

微正则系综:

$$\beta = \left[\frac{T^*(N-1)}{16 \sum_i v_i^2} \right]^{1/2}. \quad (113)$$

正则系综:

$$\beta = \left[\frac{(3N-4)kT}{\sum_i m v_i^2} \right]^{1/2}. \quad (114)$$

4. 有限差分方法

4.1 差分公式

向前、向后、中心差分 (h 为步长):

$$f(x+h) - f(x), \quad f(x) - f(x-h), \quad f(x+h) - f(x-h). \quad (115)$$

Taylor 展开:

$$f(x+h) = f(x) + f'(x)h + \frac{1}{2}f''(x)h^2 + O(h^3), \quad (116)$$

$$f(x-h) = f(x) - f'(x)h + \frac{1}{2}f''(x)h^2 + O(h^3). \quad (117)$$

因此

$$\begin{aligned} f'(x) &\approx \frac{f(x+h) - f(x)}{h}, & O(h), \\ f'(x) &\approx \frac{f(x) - f(x-h)}{h}, & O(h), \\ f'(x) &\approx \frac{f(x+h) - f(x-h)}{2h}, & O(h^2), \\ f''(x) &\approx \frac{f(x+h) + f(x-h) - 2f(x)}{h^2}, & O(h^2). \end{aligned} \quad (118)$$

4.2 常微分方程

对

$$\frac{dy}{dx} = f(x, y), \quad y(x_0) = y_0. \quad (119)$$

4.2.1 欧拉法

$$y_{n+1} = y_n + hf(x_n, y_n). \quad (120)$$

4.2.2 梯形法

$$y_{n+1} = y_n + \frac{h}{2} [f(x_n, y_n) + f(x_{n+1}, y_{n+1})]. \quad (121)$$

迭代：

$$y_{n+1}^{(0)} = y_n + hf(x_n, y_n), \quad y_{n+1}^{(k)} = y_n + \frac{h}{2} [f(x_n, y_n) + f(x_{n+1}, y_{n+1}^{(k-1)})]. \quad (122)$$

4.2.3 龙格库塔方法（二阶）

$$\begin{cases} y_{n+1} = y_n + hk_2, \\ k_1 = f(x_n, y_n), \\ k_2 = f\left(x_n + \frac{h}{2}, y_n + \frac{h}{2}k_1\right). \end{cases} \quad (123)$$

4.3 含时问题

$$\frac{\partial u}{\partial t} = a \frac{\partial^2 u}{\partial x^2} \quad (124)$$

$$u(x, t + \Delta t) = u(x, t) + a \frac{\Delta t}{(\Delta x)^2} [u(x + \Delta x, t) + u(x - \Delta x, t) - 2u(x, t)] \quad (125)$$

4.4 Poisson 方程

在直角坐标中将微分算符转换为差分：

$$\nabla^2 f = \frac{f_1 + f_2 + f_3 + f_4 - 4f_0}{h^2}. \quad (126)$$

Poisson 方程：

$$\nabla^2 \phi = g(x, y). \quad (127)$$

因此

$$\frac{\phi_1 + \phi_2 + \phi_3 + \phi_4 - 4\phi_0}{h^2} = g_0, \quad (128)$$

即

$$\phi_0 = \frac{1}{4} (\phi_1 + \phi_2 + \phi_3 + \phi_4 - h^2 g_0). \quad (129)$$

4.5 不等距公式

$$\frac{\partial \phi}{\partial x} \approx \frac{\phi_1 - \phi_0}{h_1} \approx \frac{\phi_0 - \phi_3}{h_3}. \quad (130)$$

$$\left(\frac{\partial \phi}{\partial x} \right)_0 \approx \frac{h_3^2(\phi_1 - \phi_0) + h_1^2(\phi_0 - \phi_3)}{h_1 h_3 (h_1 + h_3)}, \quad O(h^2). \quad (131)$$

$$\left(\frac{\partial^2 \phi}{\partial x^2} \right)_0 \approx 2 \cdot \frac{h_3(\phi_1 - \phi_0) - h_1(\phi_0 - \phi_3)}{h_1 h_3 (h_1 + h_3)}, \quad O(h^2). \quad (132)$$

原理： ϕ_1, ϕ_3 在 0 处 Taylor 展开。

4.6 二维 Poisson 型方程

$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + f(x, y)\phi = g(x, y). \quad (133)$$

从而

$$\frac{\phi_1 + \phi_3 - 2\phi_0}{h^2} + \frac{\phi_2 + \phi_4 - 2\phi_0}{h^2} + f_0 \phi_0 = g_0. \quad (134)$$

即

$$\phi_{i+1,j} + \phi_{i-1,j} + \phi_{i,j+1} + \phi_{i,j-1} + (h^2 f_{ij} - 4) \phi_{ij} = h^2 g_{ij}. \quad (135)$$

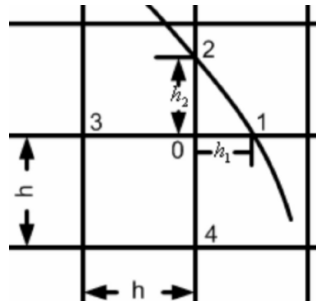
$f = 0$ 即 Poisson 方程.

4.7 边界条件

第一类边界条件：

零次插值：

$$\phi_0 \approx \phi_{\text{near}}. \quad (136)$$



线性插值：

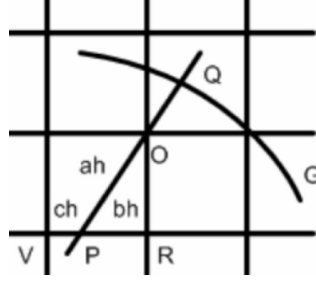
$$\phi_0 \approx \frac{h\phi_1}{h+h_1} + \frac{h_1\phi_3}{h+h_1}. \quad (137)$$

或

$$\phi_0 \approx \frac{h\phi_2}{h+h_2} + \frac{h_2\phi_4}{h+h_2}. \quad (138)$$

五点插值代入五点差分公式：

$$2 \frac{h_3(\phi_1 - \phi_0) - h_1(\phi_0 - \phi_3)}{h_1 h_3 (h_1 + h_3)} + 2 \frac{h_4(\phi_2 - \phi_0) - h_2(\phi_4 - \phi_0)}{h_2 h_4 (h_2 + h_4)} + f_0 \phi_0 = g_0. \quad (139)$$



第二、三类边界条件：

$$\frac{1}{ah}(\phi_0 - b\phi_V - c\phi_R) + \alpha\phi_0 = g. \quad (140)$$

对应

$$\left(\frac{\partial \phi}{\partial n} + \alpha \phi \right) \Big|_{\Gamma} = g. \quad (141)$$

4.8 迭代法 (Poisson 方程)

直接迭代法：

$$\phi_{ij}^{(k+1)} = \frac{1}{4} \left(\phi_{i+1,j}^{(k)} + \phi_{i-1,j}^{(k)} + \phi_{i,j+1}^{(k)} + \phi_{i,j-1}^{(k)} - h^2 g_{ij} \right). \quad (142)$$

高斯赛德尔法：

$$\phi_{ij}^{(k+1)} = \frac{1}{4} \left(\phi_{i+1,j}^{(k)} + \phi_{i-1,j}^{(k+1)} + \phi_{i,j+1}^{(k)} + \phi_{i,j-1}^{(k+1)} - h^2 g_{ij} \right). \quad (143)$$

超松弛迭代法：

$$\phi_{ij}^{(k+1)} = \phi_{ij}^{(k)} + \frac{\omega}{4} \left(\phi_{i+1,j}^{(k)} + \phi_{i-1,j}^{(k+1)} + \phi_{i,j+1}^{(k)} + \phi_{i,j-1}^{(k+1)} - h^2 g_{ij} - 4\phi_{ij}^{(k)} \right). \quad (144)$$